

Position: High Frequency Soft Magnetic Materials Scientist (PhD or MS)

Position Contact: Prof. Paul Ohodnicki, pro8@pitt.edu

Anticipated Position Start Date: Summer 2026

Position Description:

The Department of Mechanical Engineering and Materials Science (MEMS) at the University of Pittsburgh invites applications for a research associate with an emphasis on soft magnetic materials and high frequency electromagnetic property characterization. Near-term anticipated project work will include processing and characterization of ferrite, amorphous, and nanocrystalline soft magnetic materials for high frequency applications, key materials for electrification of the transportation and industrial sectors. The position is funded as a result of a number of major research programs including a [\\$3.9M program supporting next generation magnetics needs of the US Navy](#) and a [\\$2.5M program in the area of distribution transformers for the US Department of Energy](#) as well as the [Advanced Magnetism for Power and Energy Development](#) consortium based at the University of Pittsburgh. The research associate will have an opportunity to collaborate with an interdisciplinary group of undergraduate, graduate, and PhD level researchers in magnetic materials and electric power conversion applications. In addition, the research associate can expect close collaboration with other researchers from industry and national labs including NASA, Department of Energy, Department of Defense, and both large and small companies.

The advisor will be Prof. Paul Ohodnicki, and the research associate will have full access to facilities available within the Ohodnicki Lab (<https://www.engineering.pitt.edu/OhodnickiLab/>) at the University of Pittsburgh, the extensive shared facilities of the Advanced Magnetism for Power and Energy Development (AMPED, <https://www.engineering.pitt.edu/AMPED/>), and the Petersen Nanoscale Fabrication and Characterization Facility (NFCF, <https://www.tour.pitt.edu/tour/nanoscale-fabrication-and-characterization-facility>) amongst other facilities across campus.

More information about the Ohodnicki Lab research focus and interests can be found here:

<https://www.youtube.com/watch?v=7Vn7XJmHGr4&feature=youtu.be>

Successful applicants should display strong experience synthesizing and characterizing magnetic materials as well as both standard magnetometry techniques and high frequency measurements in the kHz-GHz range. Applicants should have a graduate degree (MS or PhD) in materials science and engineering, applied physics, electrical engineering, or related field. Prior experience in bulk powder and ceramic material processing and characterization, metallurgy, rapid solidification, additive manufacturing, as well as magnetometry techniques and high frequency electromagnetic property characterization is desired. Relevant prior skills and experience of interest include rapid solidification, milling, sieving, sintering, compaction, x-ray diffraction, electron microscopy, vibrating sample magnetometry, as well as impedance and vector network analyzer characterization.

Anticipated project work assignments will include characterization and synthesis of amorphous and nanocrystalline alloys as well as powdered soft magnetic materials, including the application of advanced (including additive) manufacturing techniques to synthesize relevant components for detailed characterization. In addition to conducting research, duties will also include preparing reports, performing literature reviews, supervising and mentoring students, and assisting with other projects in the laboratory. The research associate will also be responsible for overseeing scaled magnetic material synthesis laboratories currently being established at the [Energy Innovation Center](#).

Application Process:

Interested candidates should contact Prof. Paul Ohodnicki (pro8@pitt.edu) with a updated resume.